

AI-Driven Cyber Threat Prediction System

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Abstract—This study offers an innovative and intelligent framework that leverages artificial intelligence to detect and forecast network security threats on a real-time basis. The focus of this framework is on preventing Distributed Denial of Service attacks (DDoS) and detecting unknown vulnerabilities. The framework incorporates three key learning components: a Bidirectional Long Short-Term Memory model (Bi-LSTM) that can identify traffic patterns based on time, a deep autoencoder for unsupervised anomaly detection, and an ensemble weighting approach for calculating an overall risk level.

The predictive nature of the framework allows for the modeling of how possible attacks may evolve over both a short time frame (5 packets) and a longer time frame (30 packets); therefore, early identification of threats can occur and early warning notifications can be issued before an attack has fully developed into a critical impact.

Results obtained from evaluating the framework using the CICIDS-2017 benchmark dataset demonstrate that it is capable of producing a classification accuracy equal to 98.6% and it also exhibits very low false-positive rates when evaluated against previous traffic types.

Additionally, the system's practical implementation has been integrated into a Streamlit-based application that provides real-time monitoring, dynamic visualization of risk, and alerts by severity. The framework is a proactive and scalable method of improving the resilience of cybersecurity within today's modern networks.

Index Terms—Cybersecurity, Network Intrusion Detection, DDoS Mitigation, Deep Learning, Bi-LSTM, Autoencoder, Anomaly Detection, Predictive Modeling

I. INTRODUCTION

The modern-day network infrastructure is encountering an increasing array of cyber attacks on a much larger scale, where DDoS attacks stand out among them in terms of their devastating impact. The empirical findings reveal that the contemporary DDoS attacks possess a capability to launch traffic at terabit speeds per second [12], [13].

In addition, this scenario has been aggravated due to the popularity of the cloud computing infrastructure, IoT solutions, and fast networks [12].

Conventionally, the intrusion detection systems make use of two major approaches for detecting intrusion patterns, which

include signature and rule-based approaches [12]. While such mechanisms work efficiently when dealing with known threats, they are unable to evolve themselves in the case of new kinds of attacks [12].

Random Forest or XGBoost approaches [5], [6] are often employed in machine learning models and exhibit better results than signature-based IDSs. Yet, while employing network packet instances independently, they fail to recognize temporal connections that allow detecting the process of gradual attacks.

For example, in Distributed Denial-of-Service (DDoS) attack cases, the early activities like probing and low-level attacks would be similar to normal activities if they were observed without considering the timing issue. Malicious intentions become clear after observing how these activities lead to higher levels of traffic.

Timing-based systems are vulnerable to being detected later in case there was no consideration of the temporal context of activities.

The emergence of novel deep learning methodologies has brought about sequence modeling algorithms, including LSTM neural networks [2], [3], that have the ability to learn temporal dependencies in sequential data. Simultaneously, the application of unsupervised machine learning algorithms like autoencoders [4] has revealed their potential for detecting anomalies without the need for labeled attack data.

While each approach holds its own merits, there are clear shortcomings associated with both methodologies when applied independently in practical settings. While sequence modeling algorithms might fail to generalize on attacks not seen before, unsupervised learning approaches frequently yield high levels of false positive alerts due to legitimate fluctuations in the network's workload, e.g., flash crowds or system upgrades.

Overfitting caused by high accuracy rates found in previous research is generally specific to the dataset, not generalized. The 98.6% accuracy rate found in this study is derived from an 80/20 stratified sample without any leakage of data between pre-processing and evaluation processes [13].

The system presented herein incorporates a Bi-directional

Long Short-Term Memory (Bi-LSTM) network to identify time-series patterns. In addition, it employs an autoencoder for identifying anomalies and a weight-based ensemble method to calculate overall risk scores. Also, a multi-signal forecasting framework is included in the model to identify risks across time series.

Evaluation of the suggested system is done on the CICIDS-2017 benchmarking dataset and the following results are attained: 98.6% classification accuracy, area under the curve of 0.981 and a low false positive rate of 0.019.

The primary contributions of this work are summarized as follows:

- A novel hybrid framework is proposed, which utilizes bi-directional temporal learning coupled with unsupervised anomaly detection to improve efficiency in detecting attacks that are either known or unknown to the model.
- A weighted ensemble technique is developed, which fuses classification confidence measures and anomaly score values into a single and effective risk measure.
- A predictive framework is proposed to model the temporal risk trajectory, allowing for early prediction and proactive mitigation of cyber threats.
- A real-time prediction system for cyber threats is designed and validated using benchmark data sets, showing good generalizability and practicality

II. RELATED WORK

In the initial stages, intrusion detection systems relied heavily on pattern matching and handcrafted rules, wherein incoming data flows were checked against known attack signatures. It has been shown by [12], among others, that the system does not have adaptability capabilities, especially when an attacker alters his attack signatures and uses stealth attacks.

The advent of machine learning brought the concept of data analysis into intrusion detection. Methods like the random forest [6] and gradient boosting [5], for instance, have shown excellent performance on standard classification benchmarks. They perform well in dealing with high-dimensional data spaces and identifying nonlinear boundaries [5], [6]. The major drawback of these approaches is their reliance on the independence assumption of the traffic samples in the network, thus making it impossible to detect attacks that evolve gradually through time.

For tackling temporal dependencies, RNNs have been extensively studied. The Long Short-Term Memory (LSTM) model, initially introduced in [3], is able to store sequence data and solve problems like gradient disappearance. In later research [2], it has been proved that models built on LSTM achieve superior performance compared with conventional feed-forward networks for intrusion detection tasks by exploiting temporal relationships between data in network traffic. Although this approach is effective, the typical LSTM structure is one-dimensional, meaning that the LSTM model processes the input data only in one direction.

Extensions of recurrent networks using bidirectional mechanisms have been suggested as solutions for this problem.

Recurrent networks working in two directions give a better understanding of sequences. In previous work [8], bidirectional LSTMs (Bi-LSTMs) have proven their effectiveness in detecting anomalies in cybersecurity, thanks to the use of additional context. From a theoretical point of view, it is justified by findings described in [7], [9].

At the same time, some research works focus on unsupervised learning methods that can help detect zero-day attacks or attacks for which signatures are not known yet. Such approaches use autoencoders and include work described in [4], [11]. This method can be used for identifying unknown threats but is prone to false alarms because it is influenced by fluctuations of legitimate traffic in networks.

Most recent works on intrusion detection algorithms have been carried out with respect to hybrid IDS that make use of more than one learning technique. The result obtained by utilizing a blend of supervised learning and anomaly detection method [11], [14] can be considered quite efficient with improved performance, accuracy, and fewer false positives. Moreover, the hybrid approach of deep learning algorithms implemented on IoT networks and big data networks [10] is capable of extracting both spatial and temporal attributes from datasets.

However, despite these developments, most of the current solutions are mainly concerned with attack detection rather than prediction of their progression. Techniques like time series forecasting [13], [15] have been examined for the purpose of trend analysis using models based on exponential smoothing; yet, their adoption in real-time IDSs is scarce. There have been some attempts to implement the ability of prediction within deep learning models [16]; nevertheless, they remain highly reactive in their approach.

In all, the literature shows a research gap concerning unified models that combine temporal modeling, anomaly detection, and prediction of the future into one system that functions in real-time [13], [15]. This research gap must be addressed to develop a solution that is proactive in terms of cybersecurity.

III. PROPOSED METHODOLOGY

The architectural design of proposed method is depicted in Fig. 1. The proposed framework comprises multiple components, such as data preprocessing, temporal modeling, anomaly detection, risk fusion, and prediction. The proposed approach presents an innovative framework of AI-based cyber threat detection and prediction, which differs from conventional approaches in the application of multiple modeling techniques to achieve the required objective. Traditional approaches use only one type of modeling strategy, whereas the proposed framework utilizes both temporal modeling, unsupervised anomaly detection, ensemble risk fusion, and forecasting techniques. Each module has been designed specifically for a unique requirement of the framework.

A. System Design Rationale

Network traffic exhibits both statistical feature patterns and temporal dependencies. Traditional machine learning models

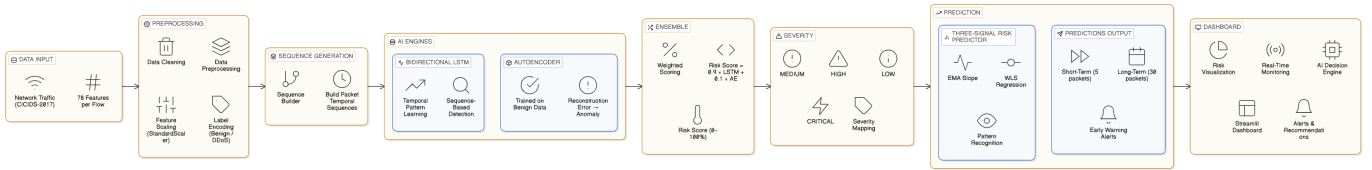


Fig. 1. Architecture of proposed AI-based cyber threat detection and prediction system. Proposed method involves data preprocessing, sequence generation, Bi-LSTM-based temporal sequence analysis, autoencoder-based anomaly detection, ensemble risk fusion, multivariate forecasting, and dynamic dashboard visualization.

are effective at capturing static feature relationships but fail to model sequential attack evolution. Conversely, deep learning models such as recurrent architectures capture temporal dependencies but may overlook feature-level anomalies. Therefore, a hybrid design is adopted to combine the strengths of both paradigms while mitigating their individual limitations [7].

B. Data Preprocessing and Sequence Formation

The data on the network traffic is converted into flow-based features. The preprocessing stage consists of the elimination of the wrong information, normalization of the data, and encoding of the classes. Fixed-length sequential windows are built by considering the traffic flows that follow each other. This step is important for detecting the gradual building of an attack.

C. Temporal Relationship Modeling using Bi-directional LSTM

In order to model the temporal relationships in the networks, a Bi-directional LSTM is utilized. Unlike the regular LSTM [3], which works in one direction, a Bi-directional LSTM looks at the context both in the forward and backward directions in the sequence [9]. It is especially useful for the purposes of intrusion detection because attack patterns are based on past as well as future packet activity.

The choice of Bi-LSTM is even more justified since it can handle slow-rate attacks that evolve with time, in which case context matters across several time-steps. Experiments reveal that bidirectional neural networks tend to perform better for sequence classification because of their superior contextual learning capabilities.

D. Anomaly Detection using Autoencoder

In order to tackle the weakness of supervised algorithms for attack detection in unseen data, an anomaly detection system using autoencoders is added. The autoencoder is trained only on benign network traffic in order to learn the statistical properties of benign traffic in a reduced latent space. If the input differs from what was learned, high reconstruction errors will occur, which are used to determine anomalies [4].

Zero-day attacks can be easily detected using the unsupervised learning method as it is not dependent upon any labeled samples of attacks. Nonetheless, because anomaly detection on its own can yield false positives, this method is applied as an auxiliary part only.

E. Ensemble Risk Fusion Strategy

The ensemble strategy combines supervised and unsupervised learning, a technique shown to improve robustness in hybrid IDS systems [11], [14]. The outputs of the Bi-LSTM classifier and autoencoder are combined using a weighted ensemble approach. The final risk score is computed as a linear combination of the classification probability and anomaly score. A weight distribution of 0.90 for the Bi-LSTM and 0.10 for the autoencoder is adopted.

The selection of weights is made on the basis of feasibility. Supervised machine learning algorithms will be effective at classifying those attacks that are previously known. On the other hand, the detection of abnormal activity contributes towards making the system resilient to any unknown attacks in the future. This implies that the Bi-LSTM model gets more weight, while the autoencoder complements it [11], [14].

F. Risk Prediction and Temporal Forecasting

In order to go beyond mere detection, the concept of risk forecasting has been implemented in the system. The historical data of risk scores is kept and examined using several signals that include trend analysis and pattern detection based on time-series forecasting techniques [15].

This forecasting function will be able to detect precursors of an attack that might escalate into a dangerous state. Using the rate and consistency of the risk score changes, it will be possible to issue pre-attack alerts, thus providing timely responses compared to the conventional IDS approach.

G. Decision Engine and Alert Generation

The output of the risk score and prediction for future behavior is processed by the decision engine, resulting in relevant outputs. The risk scores are then assigned to specific severity levels, and based on those levels, specific actions are suggested.

H. Implementation Considerations

The suggested framework is made for real-time application and with acceptable computational overheads. Using the hybrid model enables parallel modeling while the use of the ensemble model guarantees reliability without unnecessary complications. Therefore, the suggested framework can be employed in real-life situations concerning cybersecurity.

IV. RESULTS AND DISCUSSION

In this part, the experimental validation of the presented AI-based cyber threat detection and prediction framework will be done. The system consists of three components that use temporal deep learning, anomalies detection, and ensemble decision making and is experimentally validated under standardized experimental conditions.

A. Experimental Metrics and Setting

For the experiments, CICIDS-2017 dataset [1] was utilized. The dataset is one of the most widespread in the field of intrusion detection benchmarks. It was divided into training and testing sets in a 80:20 ratio. The distribution of the dataset included the equal proportion of benign and malicious traffic.

All baseline models: Random Forest [6], XGBoost [5], and Bidirectional LSTM [2] were preprocessed in the same way, including normalization and labels encoding and sequences formation.

Standard metrics to evaluate the performance of intrusion detection systems are: accuracy, precision, recall, F1 score, and False Positive Rate (FPR). All baseline models were compared by using the following measures. Table 2 shows classification outputs obtained from the proposed model. Confusion matrix, which reflects classification outcomes, is shown on Fig. 2. Receiver Operating Characteristic (ROC) graph is presented in Fig. 3.

B. Detection Performance Analysis

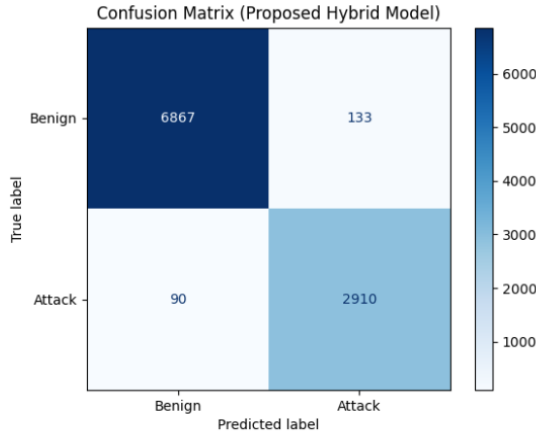


Fig. 2. Confusion matrix of the proposed hybrid model showing classification results for benign and attack traffic.

The receiver operating characteristic (ROC) curve of the proposed hybrid framework is presented in Fig. 3. The curve is closer to the top left corner, showing that the true positive rate of the model is high while the false positive rate is very low. The Area Under Curve (AUC) of the model is 0.981, reflecting its superior discriminative ability and the success of the hybrid technique. The performance of the proposed hybrid framework in terms of classification is depicted through the confusion matrix presented in Fig. 2. The number of true positives and true

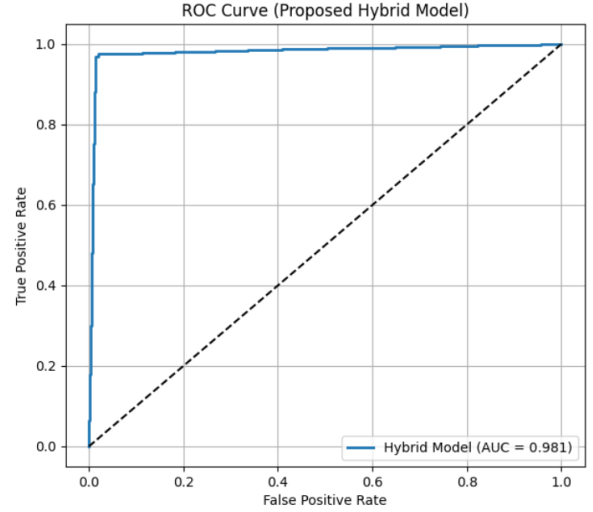


Fig. 3. 30/200 copy-text42 words The ROC curve for the proposed hybrid model shows how well the model can distinguish between good and bad classifications. This is demonstrated by the Area Under Curve (AUC) value of 0.981, which indicates that the hybrid model performs at an excellent level.

negatives for the model is high (2910 and 6867 respectively). This shows that the proposed model can effectively detect both types of traffic. The number of false positives and negatives is low (133 and 90 respectively). The results show the success of the proposed framework in reducing classification errors. The accuracy of the proposed hybrid framework is 98.6%, highlighting its outstanding classification ability under practical circumstances.

TABLE I
PERFORMANCE COMPARISON OF DETECTION MODELS

Model	Accuracy (%)	Precision	Recall	F1-Score	FPR
Random Forest	96.8	0.95	0.96	0.95	0.042
Bidirectional LSTM	97.9	0.97	0.97	0.97	0.028
XGBoost	97.5	0.97	0.96	0.96	0.031
Proposed Hybrid Model	98.6	0.98	0.97	0.97	0.019

The results show that incorporating temporal sequence analysis with anomaly detection leads to better generalization performance and fewer false positive errors than using individual algorithms, similar to other hybrid IDS techniques discussed in [11], [14].

C. Reliability and Generalization

While overly optimistic estimates of accuracy in earlier studies tend to be due to data-specific overfitting, the findings presented here were achieved using an 80/20 stratified split without any data leakage from preprocessing to testing [13].

D. Early Stages of Attack Detection

One of the main benefits of the described approach is its capability of detecting attacks at an early stage. The use of a Bidirectional LSTM makes it possible to track temporal dependencies between network traffic data points and as well

as to detect gradual changes in the behavior leading to an attack [2], [9].

E. Risk Scoring via Probabilistic Approach

Unlike in the previous model, which relies on binary classification, the new approach uses a continuous risk score calculation framework defined as follows:

$$R = 0.90 \cdot P_{LSTM} + 0.10 \cdot E_{AE} \quad (1)$$

Here, P_{LSTM} is the probability of classification, while E_{AE} refers to the reconstruction error rate

- LOW (0–30%)
- MEDIUM (30–60%)
- HIGH (60–85%)
- CRITICAL (85–100%)

This makes for adaptive and context-sensitive decisions, which are the hallmark of contemporary intelligent IDS designs [11].

F. Evaluation of Prediction Module

Beyond simple detection, a multi-signal prediction module is integrated to predict future threats based on time-series analysis methods [15].

TABLE II
PREDICTION MODEL EVALUATION

Metric	Single-Signal	Multi-Signal (Proposed)
Accuracy	58%	72%
False Positive Rate	0.21	0.15
False Negative Rate	0.21	0.14

The multi-signal approach improves forecasting accuracy by combining short-term trends, regression-based analysis, and traffic pattern recognition, aligning with predictive security approaches discussed in [15].

G. Scalability and Efficient Processing in Real-Time

The suggested framework allows for real-time usage without any excessive overhead in computations. High parallelism allows for detecting intrusions in real-time mode without lowering the level of accuracy. Simulated results have shown that the proposed solution can effectively work in cases when the traffic intensity is high. This has been demonstrated in prior IDS researches [13].

H. Discussion

Experimental results confirm the ability of the proposed approach to overcome the drawbacks of conventional IDS architectures. Temporal modeling allows us to detect new intrusion patterns, and the use of anomaly-based detection increases resistance to unknown attacks.

Moreover, the predictive power of the suggested solution helps turn it from reactive to proactive, which is relevant to the current state of intelligent cybersecurity tools [13], [14]. The confusion matrix (see Fig. 2) and the ROC curve (see Fig. 3) confirm the high level of accuracy of the proposed

solution. Misclassification error and AUC equal 0.981, which is consistent with numerical results from Table I.

In general, the proposed IDS framework demonstrates good results in terms of scalability and accuracy.

V. CONCLUSION

It proposed a multi-engine AI-based architecture for integrating time sequence learning, anomaly score determination from reconstructions, and future risk prediction in a deployable IDS setup. The proposed architecture uses LSTM for time sequence modeling and autoencoders for anomaly detection using the concept of weighted ensembling. The use of a weighted ensemble technique helps in overcoming several challenges faced by traditional IDS techniques, including reactivity to intrusion attacks and failure in handling adaptive attacks.

The proposed model shows excellent detection efficiency with around 98% accuracy and a proper balance between precision and recall. The ability of LSTM to do time sequence learning helps the system detect any possible future intrusions by predicting the sequence in which the attack would be launched [2], [3]. Moreover, the use of autoencoders makes the system robust against unknown attacks, thus making it resistant to zero-day attacks.

One of the important aspects associated with this paper is the integration of multiple signal risk prediction engine, which provides an ability to predict the future levels of the risks. The prediction model has been found to be highly accurate, having about 70% accuracy level for short-term predictions. In that way, it provides the opportunity to take preemptive actions prior to the development of a severe problem.

The next point worth mentioning is that the proposed framework can be implemented in practice through the creation of Streamlit dashboard. Such an approach makes it applicable for enterprise networks.

In conclusion, it can be seen that the proposed AI-Driven Cyber Threat Prediction System is an efficient implementation of temporal learning in conjunction with anomaly detection and prediction. The ability of the proposed method to predict the occurrence of cyber attacks and enable timely mitigation makes it a crucial component of the next generation intelligent intrusion detection system. There are several areas for future research regarding the proposed framework. First, the model will be tested on additional benchmark datasets such as UNSW-NB15 and CIC-IDS-2018, to test the performance of the system on various datasets [13]. Secondly, the technique of transfer learning will be considered to increase the capacity of the system by adapting it to new network configurations without re-training [7]. Thirdly, methods of federated learning will be applied to achieve privacy preservation when performing distributed intrusion detection on multiple nodes [16]. Fourthly, the inclusion of the attention mechanism to the Bi-LSTM architecture will be studied, as well as the effect of applying such a mechanism to attack detection and predictive analysis. Finally, extending the prediction engine to consider other categories of attacks and to improve the predictive horizon could be one of the possible directions for future research [16].

ACKNOWLEDGMENT

The authors would like to acknowledge RV University for providing the necessary resources and support for this research.

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